

CHARACTERIZING THE COLONIZATION AND POTENTIAL EXTINCTION RISK OF 12 SPECIES OF BUTTERFLIES IN CATALONIA

Author: Joana Violeta Dekker

Grade: Ciències ambientals

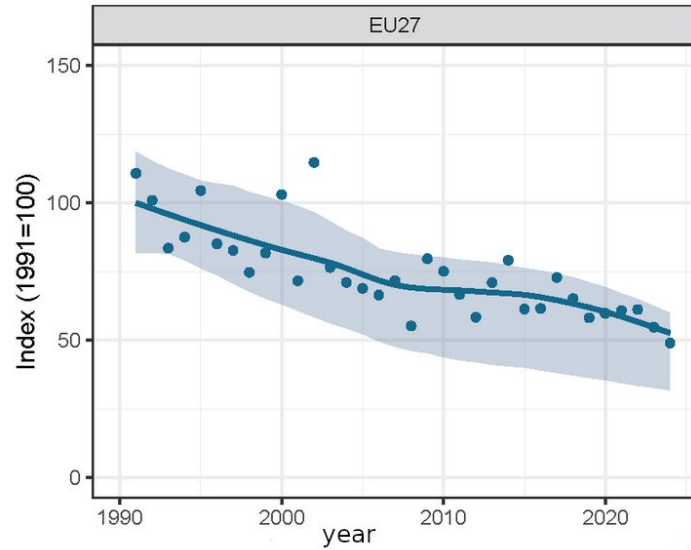
Tutor: Nuria Roura Pascal

Director and Co-director: David Alonso i Vicente J. Ontiveros

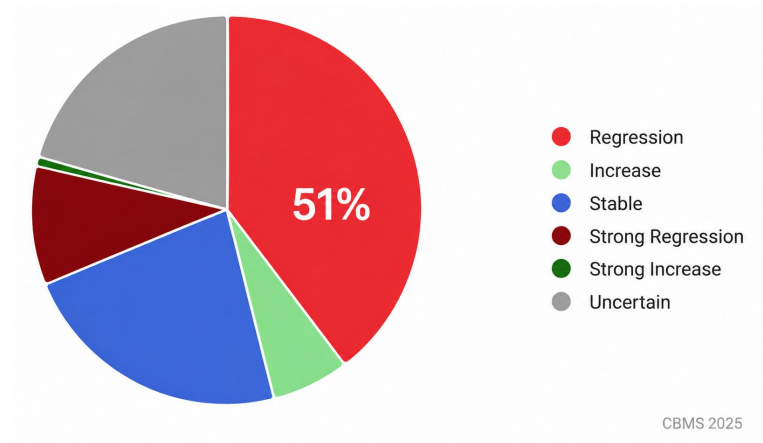
Center: CEAB-CSIC

Introduction

- Insects play a crucial role in many ecological functions.



Source: eBMS, 2026



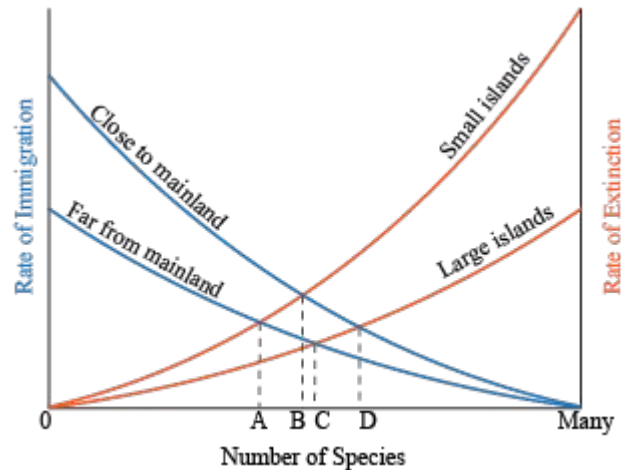
Source: CBMS, 2025

Introduction

The Theory of Island Biogeography (1967)

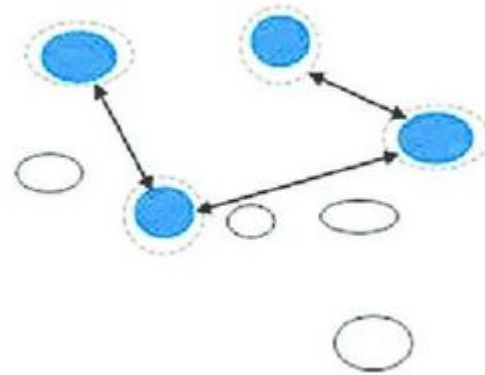
Fundamental assumptions:

- Species independence
- Species equivalence



The Metapopulation Theory (1969, 1991)

- Habitat occurs in discrete patches
- Dispersal occurs among all patches
- All populations have a substantial risk of extinction



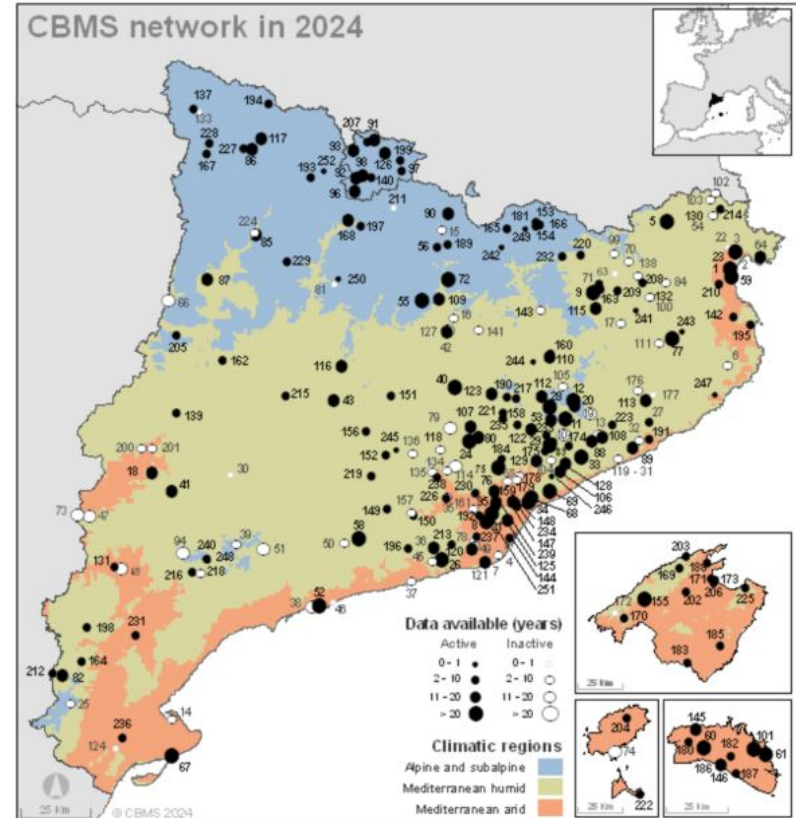
Objectives

1. Characterizing the species by a pair of parameters: the extinction and the (re-)colonization taxes
2. Compare the proposed characterization with alternative approaches based on species dispersal ability and a habitat specialization index (SSI)
3. Quantify and compare extinction risk among species by using a novel quantitative metric

Methodology

- 12 sp from more generalist to specialist and from very mobile to very sedentary species.
- The CBMS transects are grouped in three bioclimatic regions in Catalonia.
- Observed occupancy

$$p^{(o)} = \frac{n}{M}$$



Source: CBMS, 2024

Methodology

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$$p^{(o)} = \frac{n}{M}$$

ID	1994	1997	2001	2005
1	0	1	1	0
2	1	0	0	1
3	0	1	0	1
4	0	0	1	0
5	?	1	1	0
6	?	?	0	1
7	?	?	0	1
8	?	?	1	0
9	?	?	0	1
10	?	?	?	0
12	?	1	?	1
13	?	0	?	1
14	?	?	?	0
15	1	0	1	?

Source: Original work

Occupancy temporal evolution

$$\frac{dP(1; t)}{dt} = \text{COLONIZATION} - \text{EXTINCTION}$$

Occupancy temporal evolution

$$\frac{dP(1; t)}{dt} = c(1 - P(1, t)) - e P(1, t)$$

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APPLICATION



Colonization and extinction rates estimated from temporal dynamics of ecological communities: The `island` R package

Vicente J. Ontiveros^{1,2} | José A. Capitán^{1,3} | Rohan Arthur^{4,5} | Emilio O. Casamayor² | David Alonso¹

https://github.com/vankampen92/PROJECT_JOANA_TFG

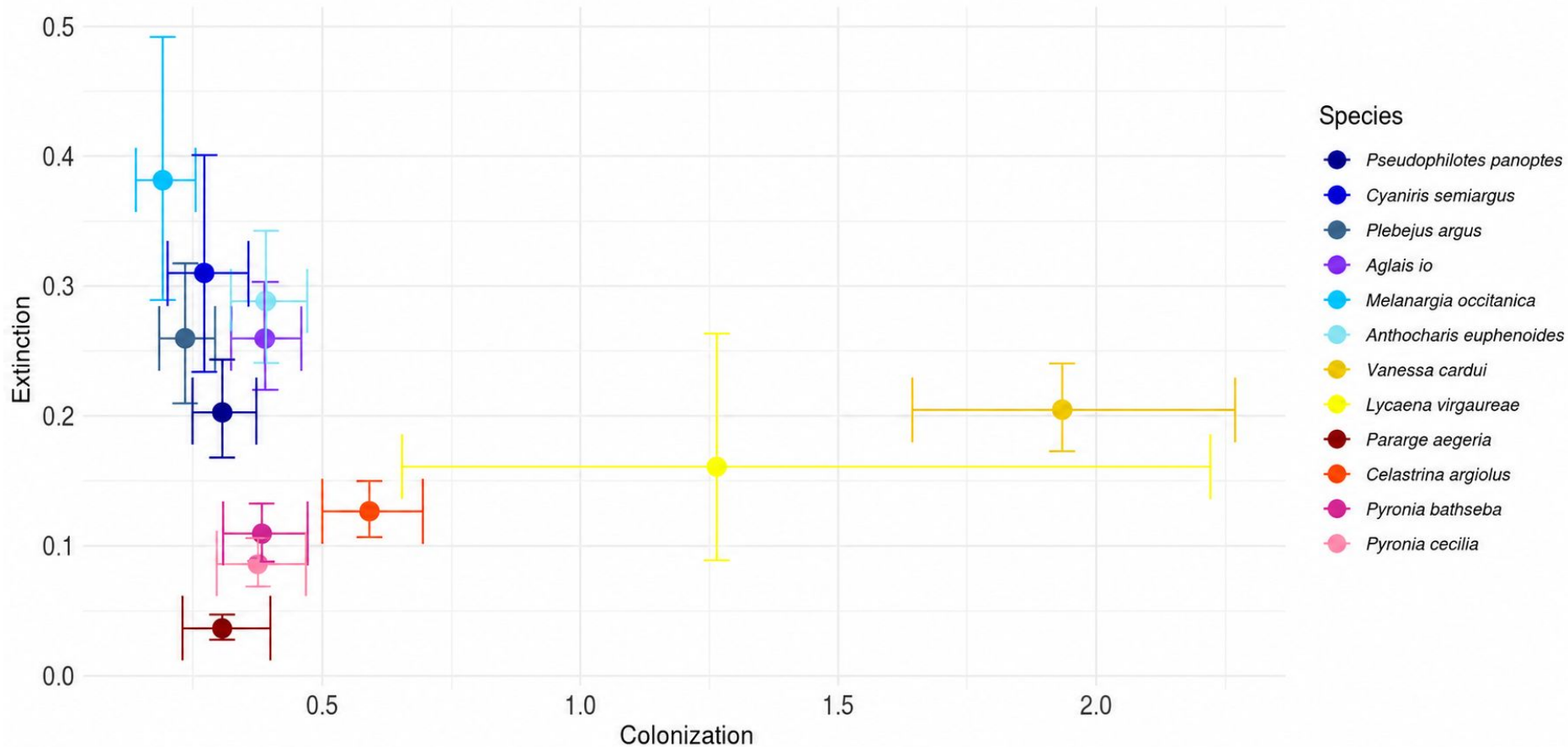
Results and discussion

1. Colonization - extinction characterization
2. Temporal trends in occupancy
3. Bioclimatic Preferences over time
4. SSI and mobility Indexs vs. colonization and extinction rates
5. Characteristic time and local extinctions

Results and discussion

1. **Colonization - extinction characterization**
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1. Colonization - extinction characterization



1. Colonization - extinction characterization

Espèces	$c [c_0, c_1] \text{ (any}^{-1}\text{)}$	$e [e_0, e_1] \text{ (any}^{-1}\text{)}$	M	ΔAIC	Ratio	p-valor
<i>Pseudophilotes panoptes</i>	0.307 [0.249, 0.373]	0.203 [0.167, 0.243]	87	-22.97	0.487	0.500
<i>Cyaniris semiargus</i>	0.271 [0.200, 0.359]	0.309 [0.233, 0.401]	35	/	0.531	0.508
<i>Plebejus argus</i>	0.234 [0.184, 0.293]	0.259 [0.209, 0.317]	66	-6.18	0.504	0.600
<i>Aglaïs io</i>	0.388 [0.325, 0.460]	0.259 [0.219, 0.303]	101	-15.36	0.532	0.442
<i>Melanargia occitanica</i>	0.191 [0.139, 0.254]	0.381 [0.289, 0.492]	36	0.93	0.503	0.696
<i>Anthocharis euphenoides</i>	0.392 [0.323, 0.471]	0.288 [0.240, 0.342]	81	-2.5	0.554	0.536
<i>Vanessa cardui</i>	1.934 [1.644, 2.270]	0.204 [0.172, 0.240]	171	-6.71	0.309	0.480
<i>Lycaena virgaureae</i>	1.264 [0.655, 2.221]	0.160, [0.089, 0.263]	20	/	0.316	0.452
<i>Pararge aegeria</i>	0.306 [0.230, 0.400]	0.036 [0.027, 0.047]	170	6.9	0.183	0.506
<i>Celastrina argiolus</i>	0.592 [0.499, 0.695]	0.126 [0.106, 0.149]	165	-59.54	0.374	0.480
<i>Pyronia bathseba</i>	0.384 [0.308, 0.472]	0.109 [0.088, 0.132]	114	-38.30	0.363	0.438
<i>Pyronia cecilia</i>	0.375 [0.296, 0.468]	0.086 [0.069, 0.106]	123	-8.03	0.319	0.520

1. Colonization - extinction characterization

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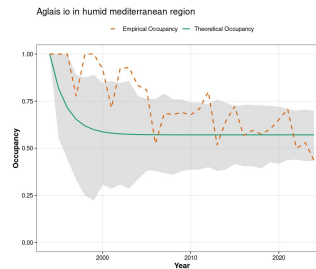
$$AIC_1 = 2(2) + 2NLL_1 = 4 + 2NLL_1$$

$$AIC_2 = 2(6) + 2NLL_2 = 12 + 2NLL_2$$

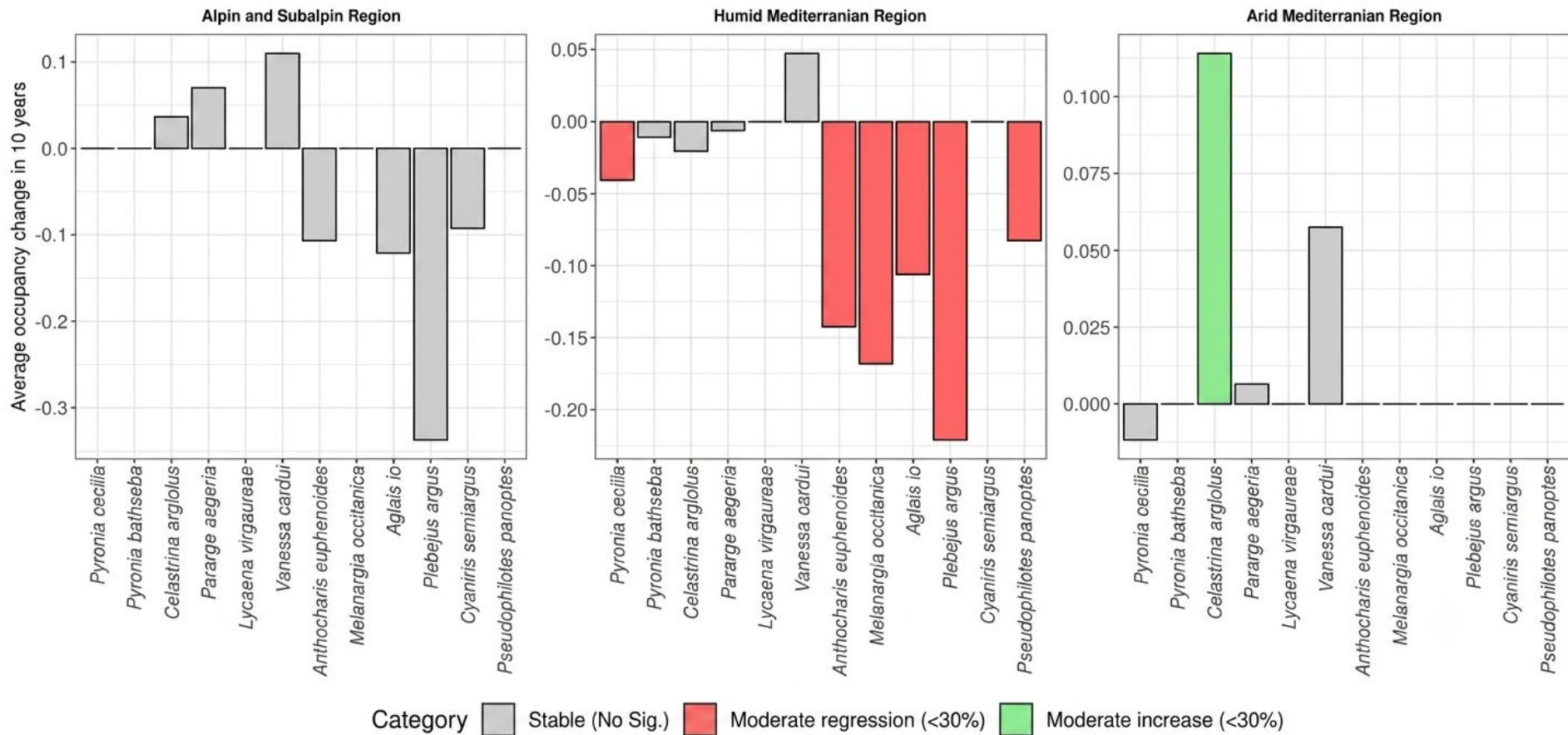
$$\Delta AIC = AIC_2 - AIC_1$$

Results

1. Colonization - extinction characterization
- 2. Temporal trends in occupancy**
3. Bioclimatic Preferences over time
4. SSI and mobility Indexs vs. colonization and extinction rates
5. Characteristic time and local extinctions

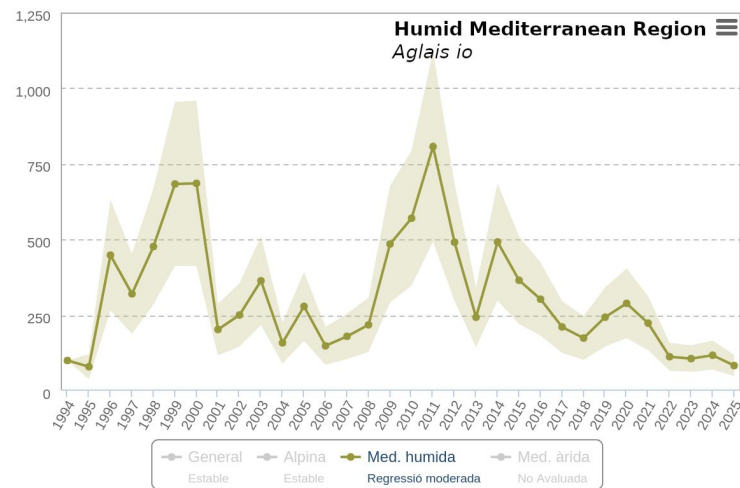
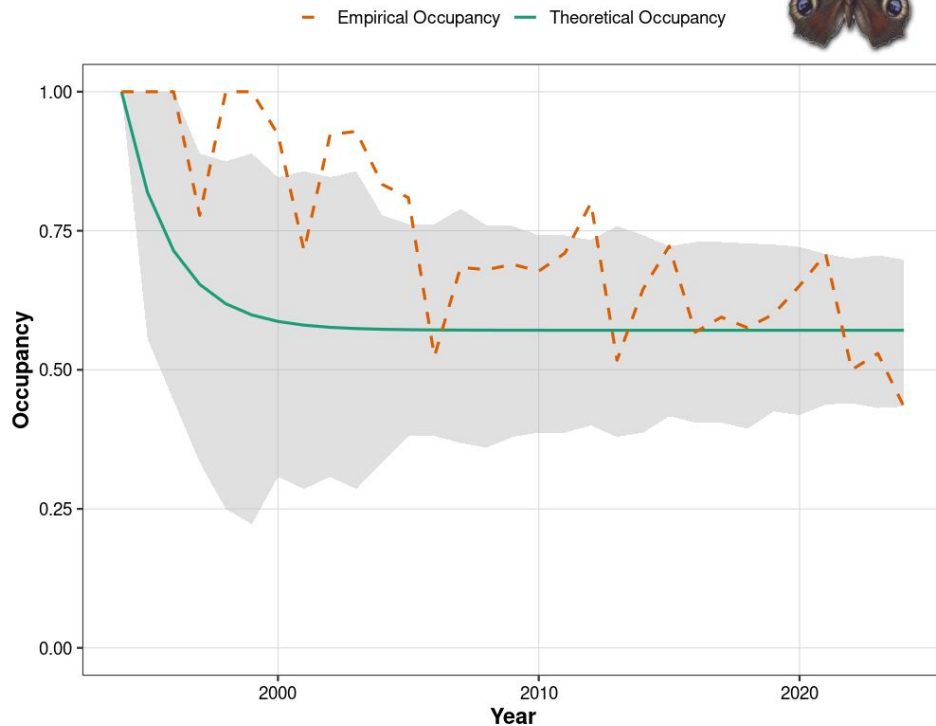


2. Temporal trends in occupancy

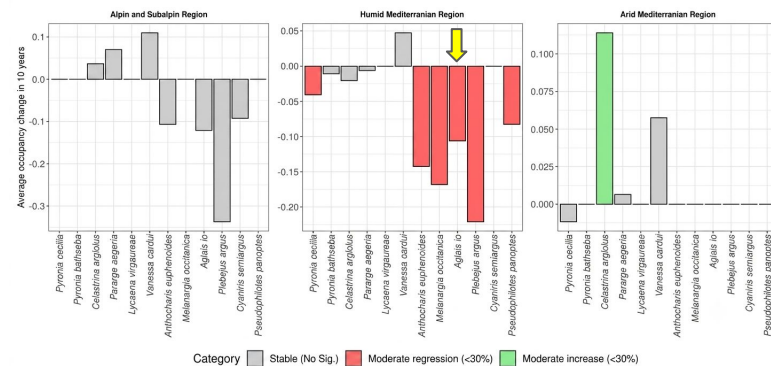


2. Temporal trends in occupancy

Aglais io in humid mediterranean region



CBMS 2025



Species	Trend	Alpine and Subalpine	Humid Mediterranean	Arid Mediterranean
<i>Pseudophilotes panoptes</i>	Abundancy	Uncertain	M.R	N.E
	Occupancy	N.E	M.R	N.E
<i>Cyaniris semiargus</i>	Abundancy	M.R	S.R	N.E
	Occupancy	S	N.E	N.E
<i>Plebejus argus</i>	Abundancy	S.R	S.R	M.I
	Occupancy	S	M.R	N.E
<i>Aglais io</i>	Abundancy	S	M.R	N.E
	Occupancy	S	M.R	N.E
<i>Melanargia occitanica</i>	Abundancy	N.E	S.R	N.E
	Occupancy	N.E	M.R	N.E
<i>Anthocharis euphenoides</i>	Abundancy	M.R	S	Uncertain
	Occupancy	S	M.R	N.E
<i>Vanessa cardui</i>	Abundancy	S	M.R	S
	Occupancy	S	S	S
<i>Lycaena virgaureae</i>	Abundancy	S	N.E	N.E
	Occupancy	N.E	N.E	N.E
<i>Pararge aegeria</i>	Abundancy	S	M.R	S
	Occupancy	S	S	S
<i>Celastrina argiolus</i>	Abundancy	M.I	S	S
	Occupancy	S	S	M.I
<i>Pyronia bathseba</i>	Abundancy	M.R	M.R	Uncertain
	Occupancy	N.E	S	N.E
<i>Pyronia cecilia</i>	Abundancy	N.E	S.R	M.R
	Occupancy	N.E	M.R	S

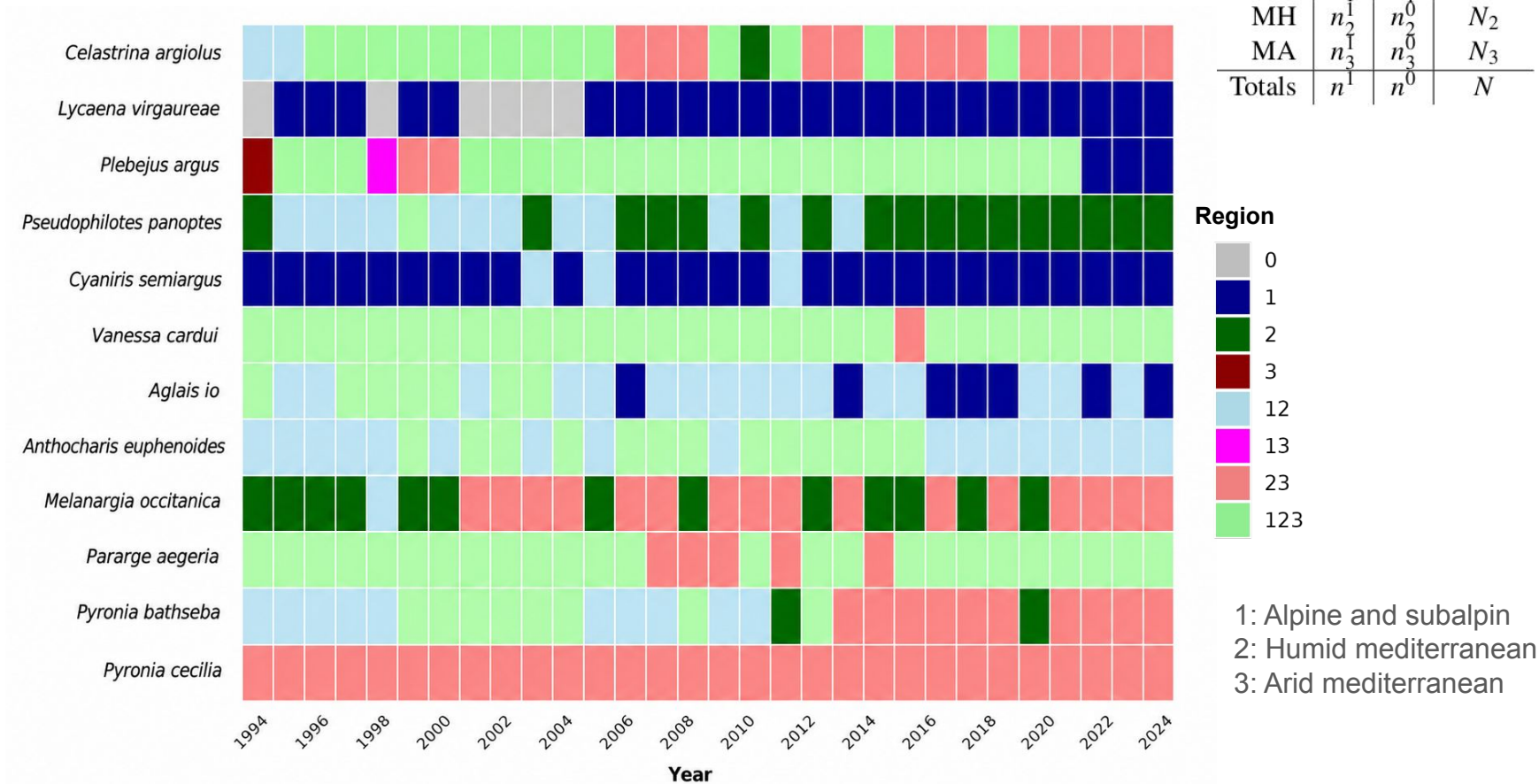
N.E: Non evaluated
M.I: Moderate increase
S: Stable
M.R: Moderate regression
S.R: Strong regression

Species	Trend	Alpine and Subalpine	Humid Mediterranean	Arid Mediterranean
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	Occupancy	S	N.E	N.E
<i>Plebejus argus</i>	Abundancy	S.R	S.R	M.I
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	Occupancy	N.E	S	N.E
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	Occupancy	N.E	M.R	S

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4. SSI and mobility Indexs vs. colonization and extinction rates
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3. Bioclimatic preferences over time



	1	0	Totals
RA	n_1^1	n_1^0	N_1
MH	n_2^1	n_2^0	N_2
MA	n_3^1	n_3^0	N_3
Totals	n^1	n^0	N

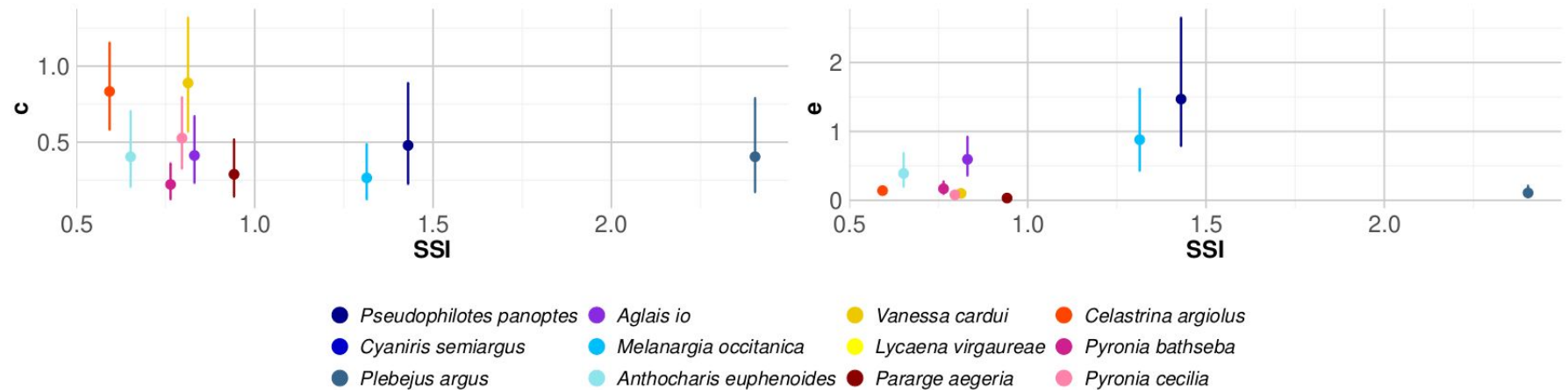
1: Alpine and subalpin
2: Humid mediterranean
3: Arid mediterranean

1. Colonization - extinction characterization
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4. SSI and mobility Indexs vs. colonization and extinction rates

- Colonization decreases with habitat specialization
- Extinction increases with habitat specialization

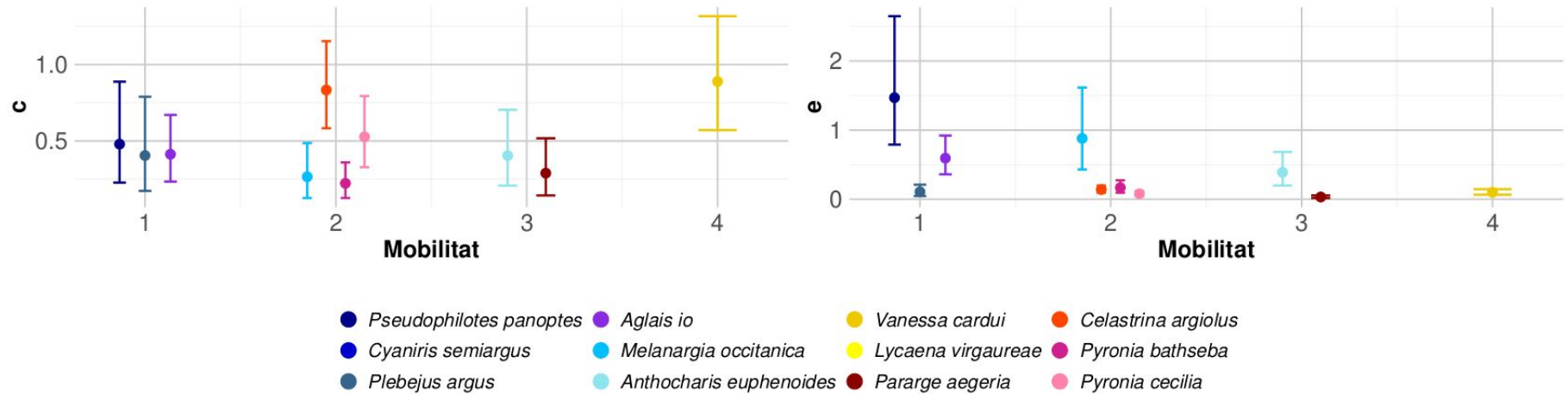
Arid mediterranean region



4. SSI and **mobility** Indexes vs. colonization and extinction rates

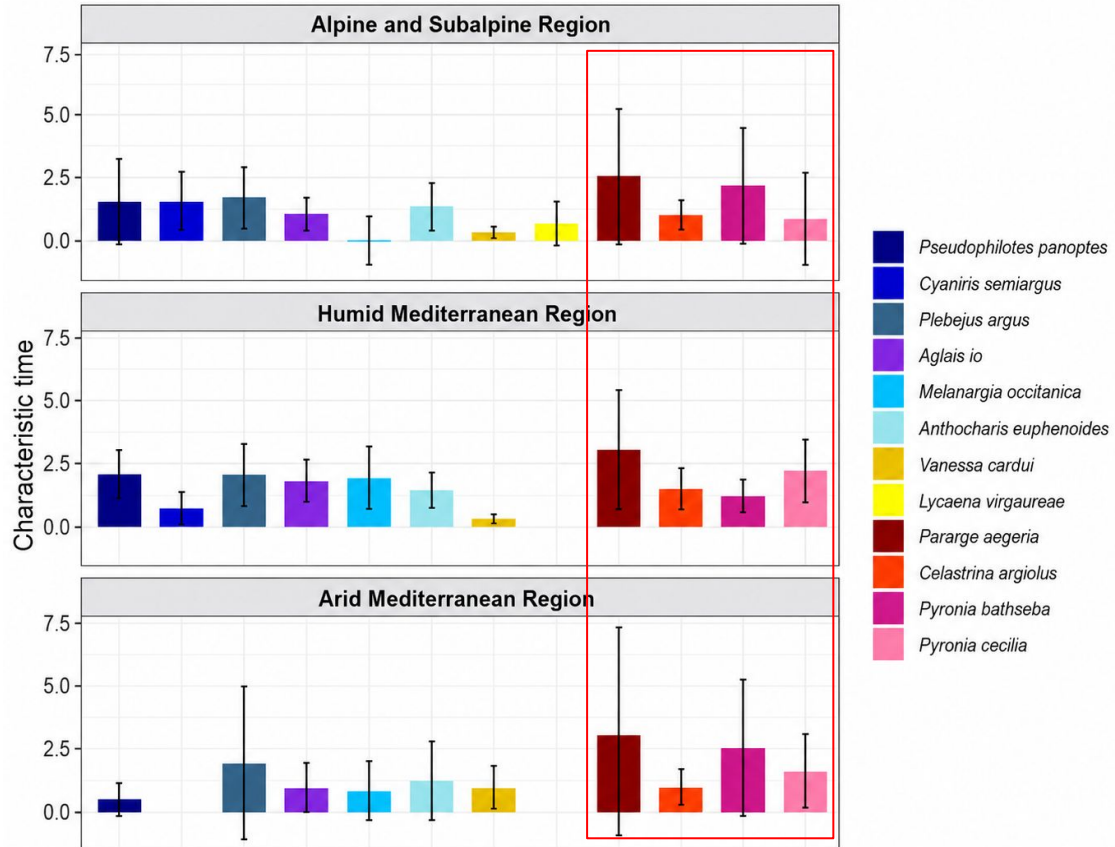
- Colonization increases with mobility
- Extinction decreases with mobility

Arid mediterranean region



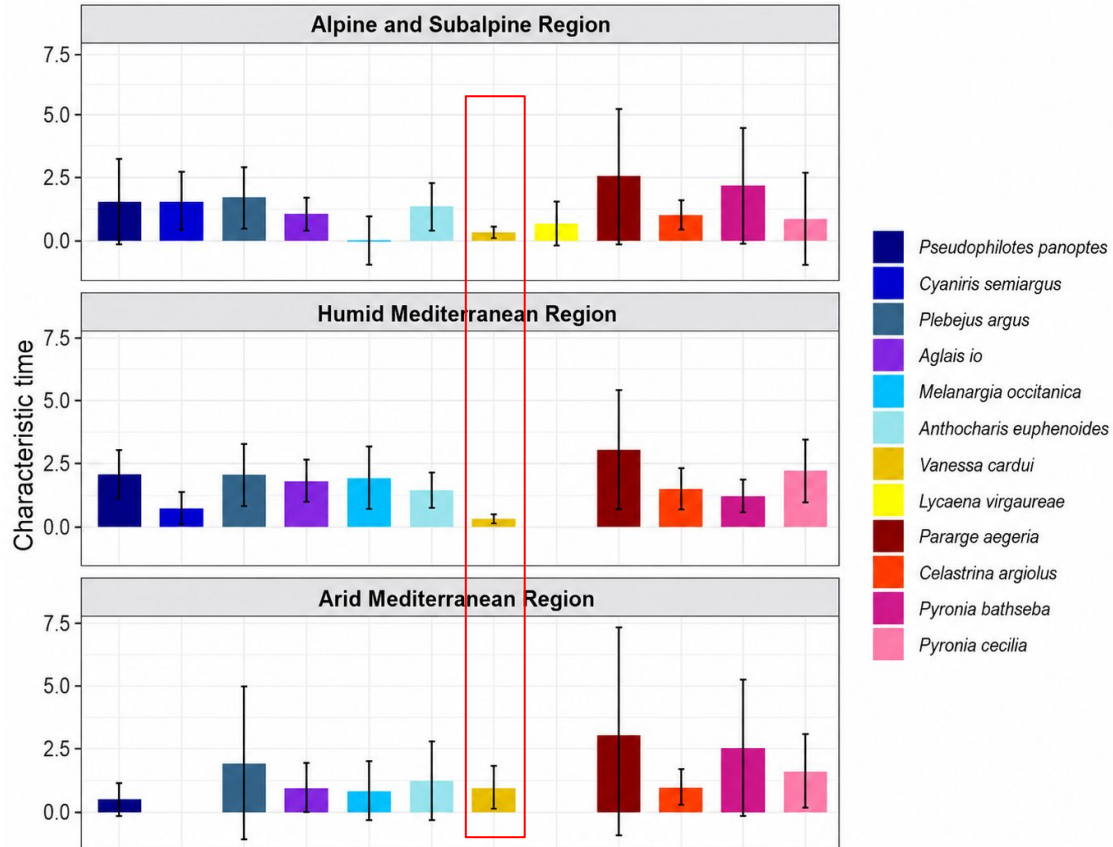
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5. Characteristic time and local extinctions



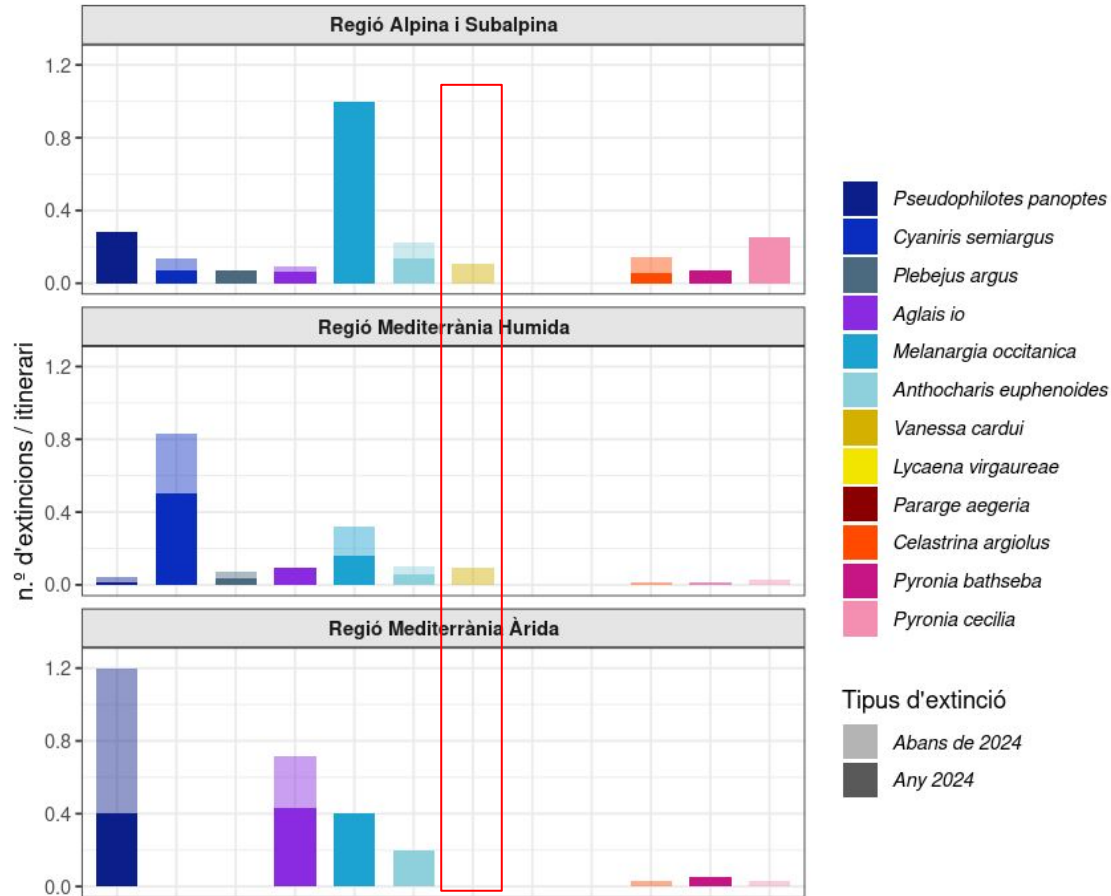
$$T = \frac{1}{c + e}$$

5. Characteristic time and local extinctions



$$T = \frac{1}{c + e}$$

5. Characteristic time and **local extinctions**



$$T = \frac{1}{c + e}$$

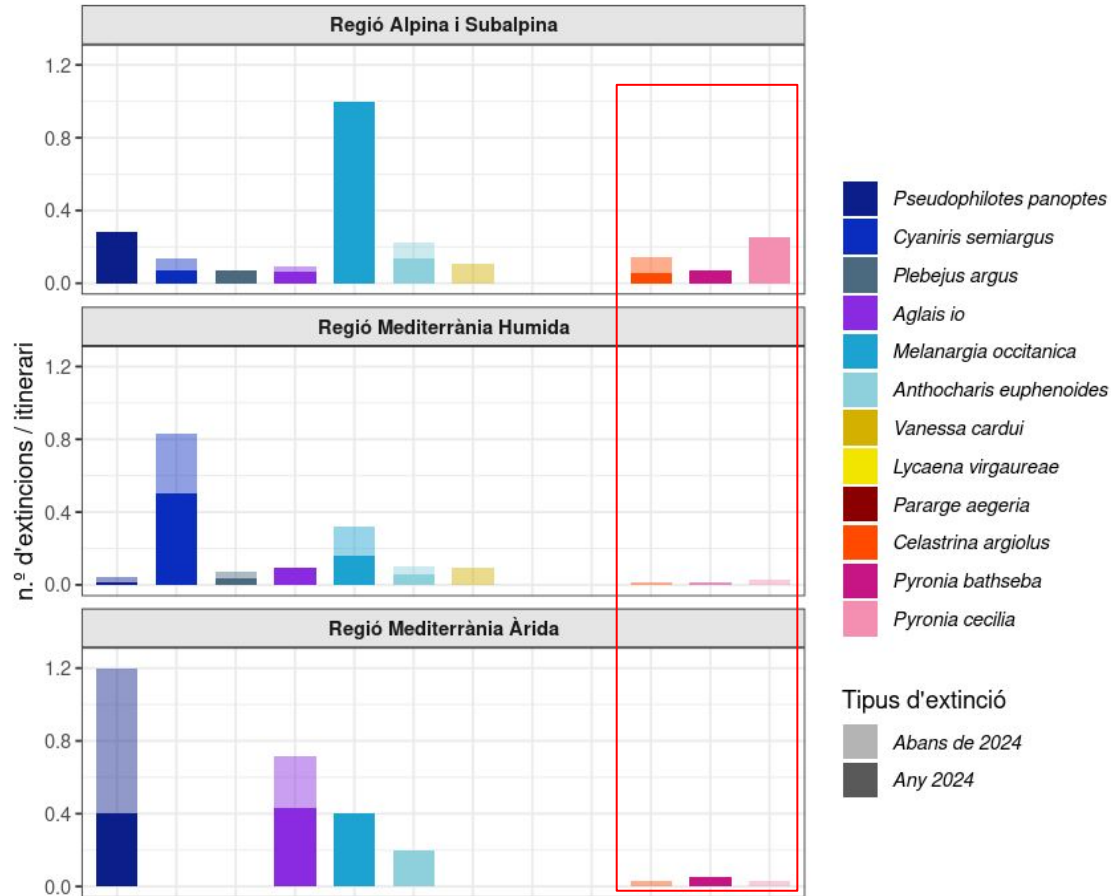
$$T(\textit{Vanessa cardui}) = 0.33 \text{ yr.}$$

$$n = 4 T = 1.32 \approx 2$$

$$\text{IT: } 0 \ 1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1 \ 2$$

Ontiveros, V. J., Capitán, J. A., Casamayor, E. O., & Alonso, D. (2021). The characteristic time of ecological communities. *Ecology*, 102(2), e03247.

5. Characteristic time and **local extinctions**



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Ontiveros, V. J., Capitán, J. A., Casamayor, E. O., & Alonso, D. (2021). The characteristic time of ecological communities. *Ecology*, 102(2), e03247.

Conclusions

1. This study shows that the annual colonization and extinction dynamics of the 12 species can be correctly characterized by a pair of parameters (c , e) using a simple model derived from the theory of island biogeography.
 - 1.1. However, the bioclimatic region is a determining factor in the dynamics of the species since most of them need 6 different parameters.
 - 1.2. In addition, the species can be classified into three different groups: species with high extinction risk, relatively stable species and an intermediate group with singular dynamics (high volatility).
2. The relationship between mobility, (habitat) specialization and colonization-extinction rates suggests consistent ecological patterns, although the analysis will need to be extended to more species to confirm them with sufficient statistical power.
3. This study presents a method to quantify occupancy trends and local extinctions, with potential applications for evaluating the extinction risk (conservation status) of Catalan butterfly species.
 - 3.1. Our methods detect significant range contractions in many species in the HM region (declining occupancies).
 - 3.2. Local extinctions can be quantified with a standardized metric that calculates the number of extinctions per transect (location).



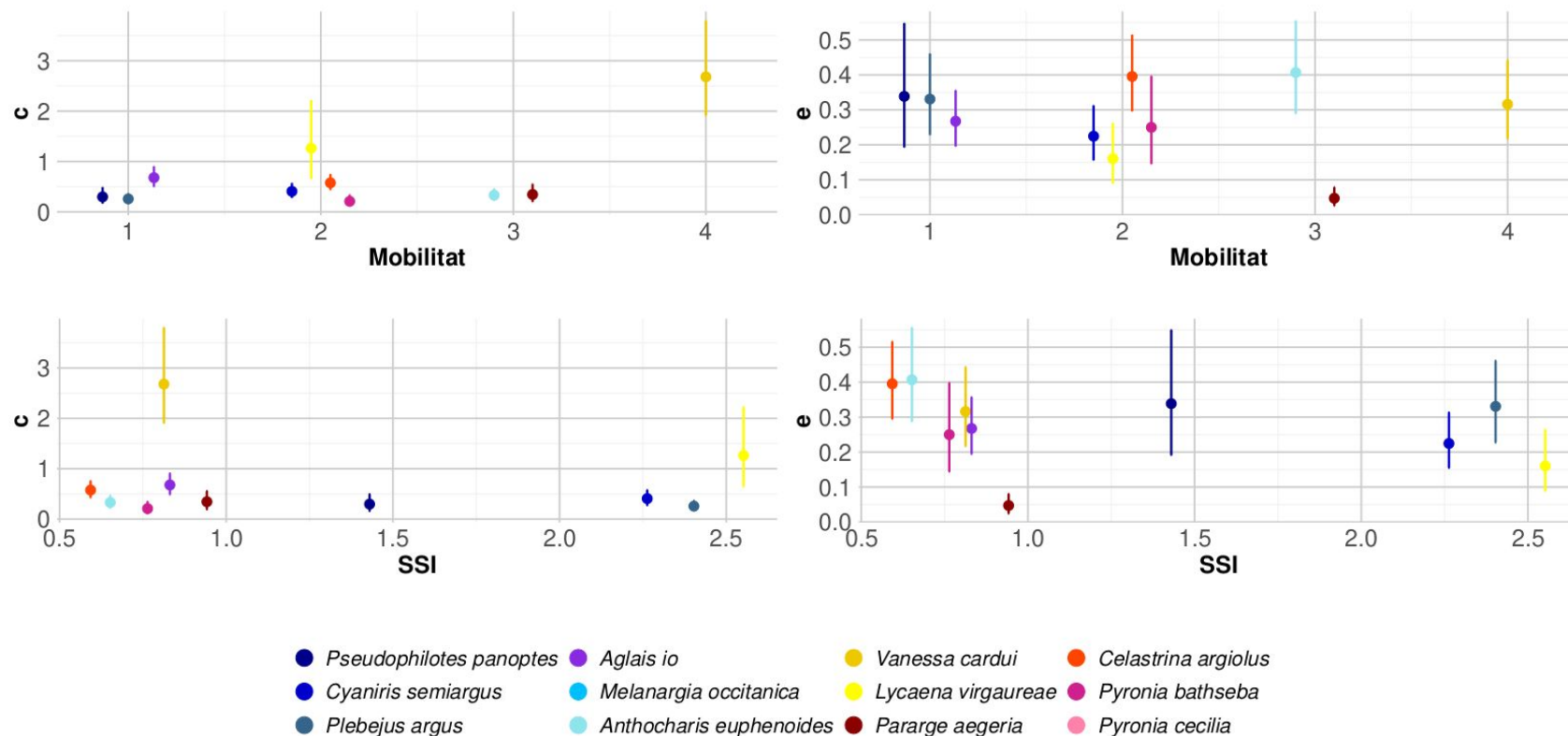
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DE GRANOLLERS



THANK YOU

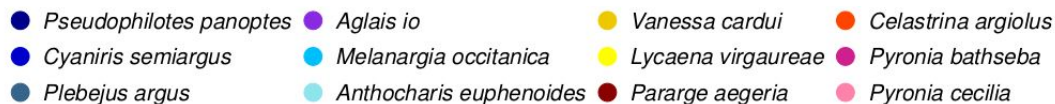
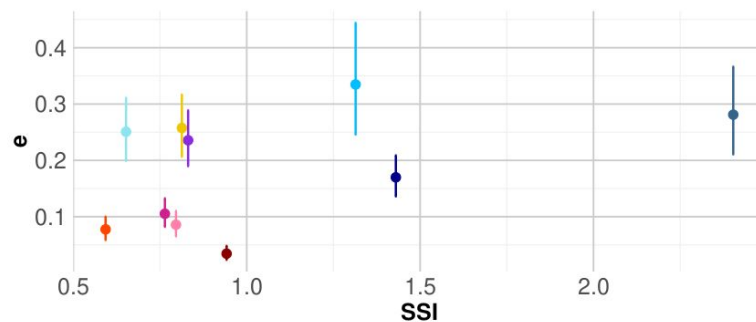
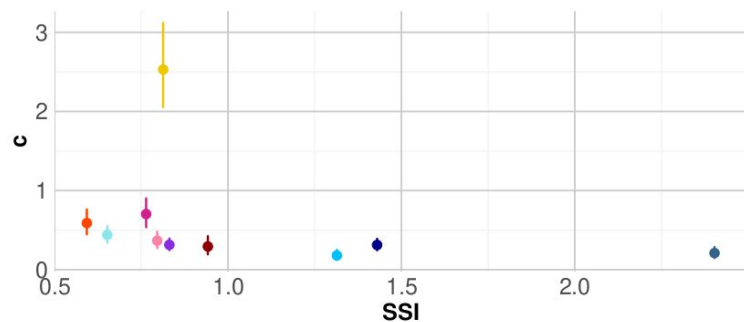
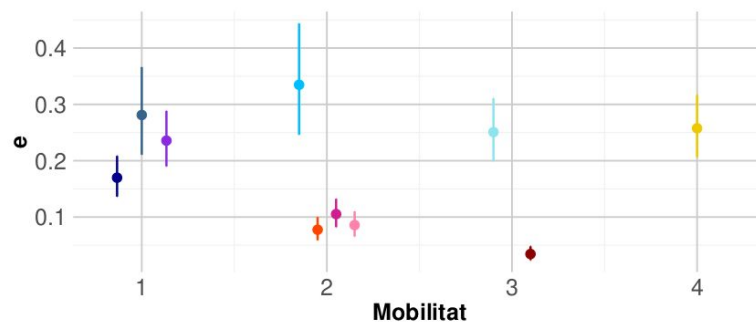
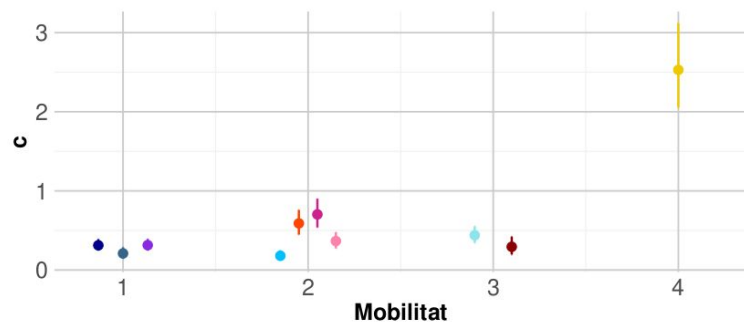
4. SSI and mobility Indexs vs. colonization and extinction rates

Alpine and subalpine region



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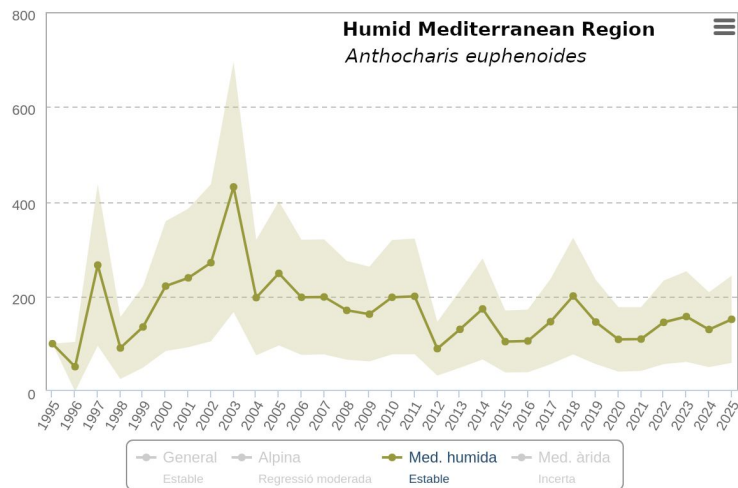
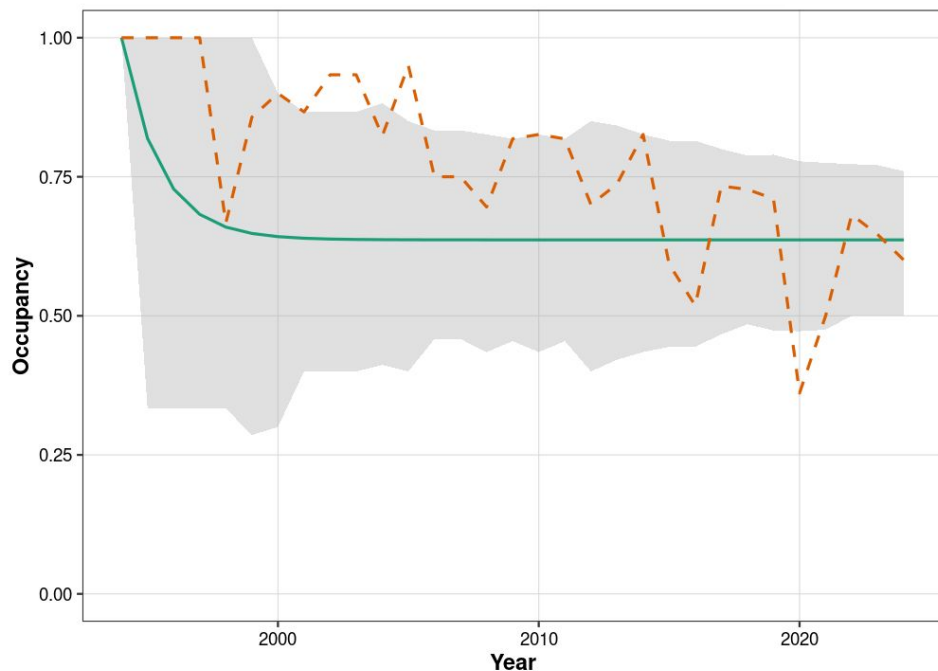
Humid mediterranean region



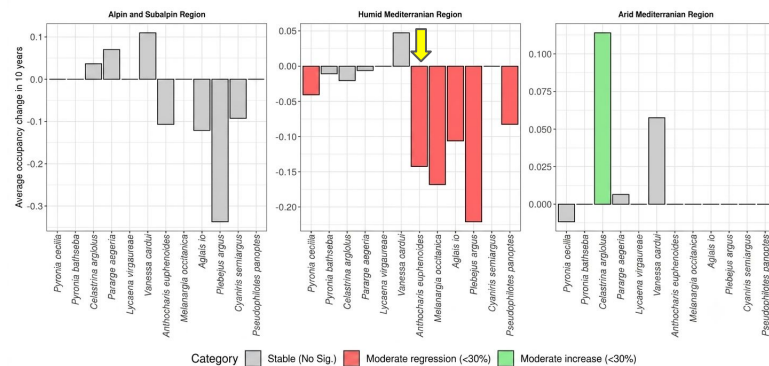
2. Temporal trends in occupancy

Anthocharis euphenoides in humid mediterranean region

— Empirical Occupancy — Theoretical Occupancy

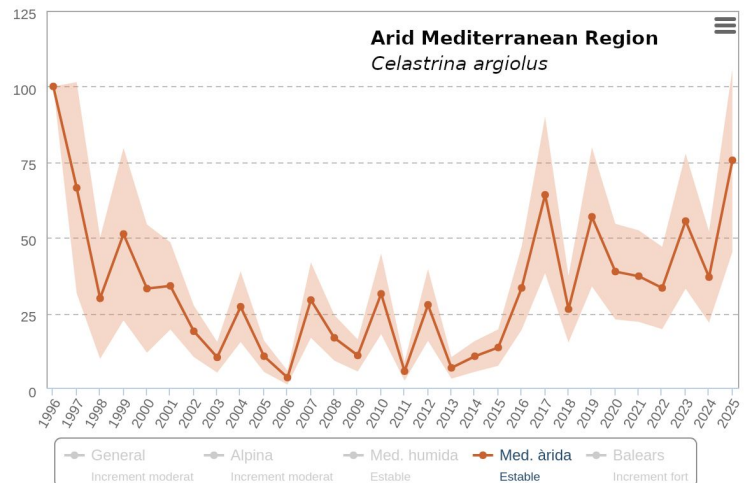
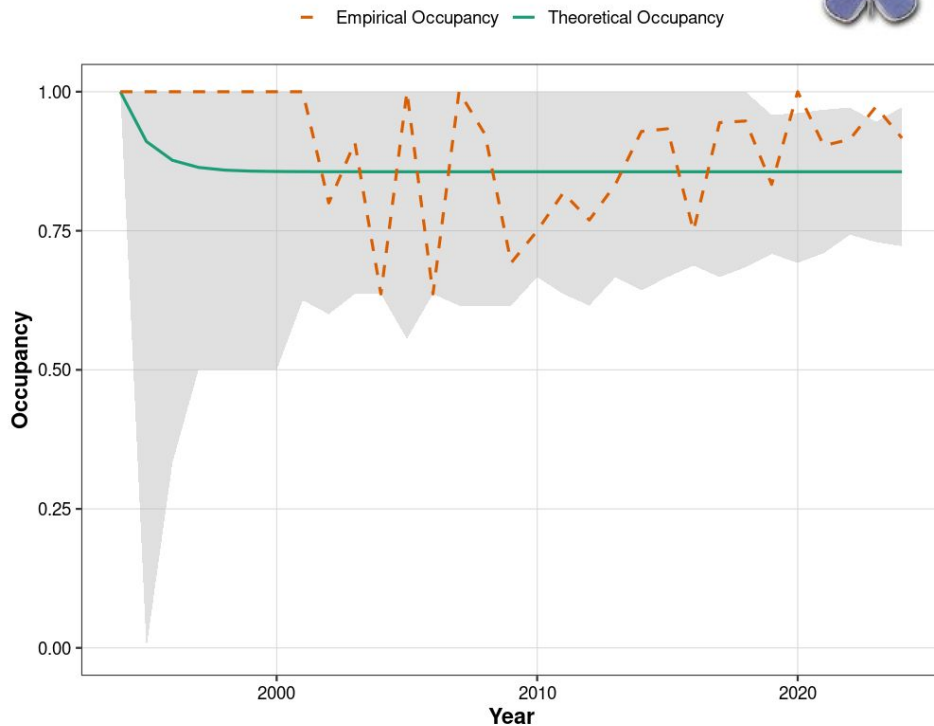


CBMS 2025

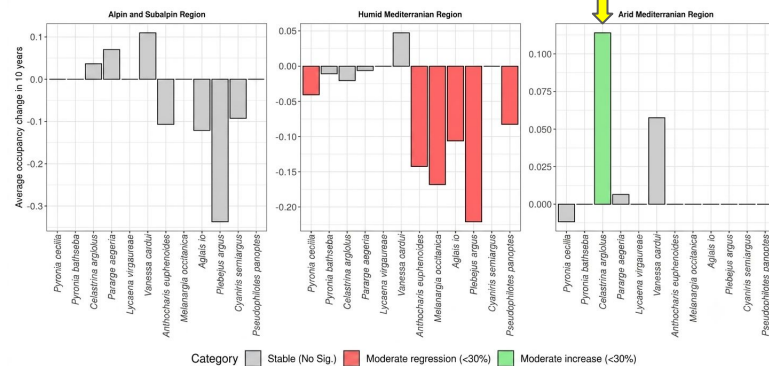


2. Temporal trends in occupancy

Celastrina Argiolus in arid mediterranean region

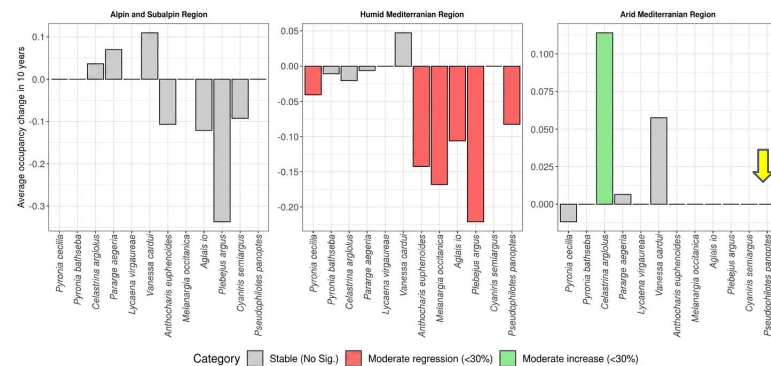
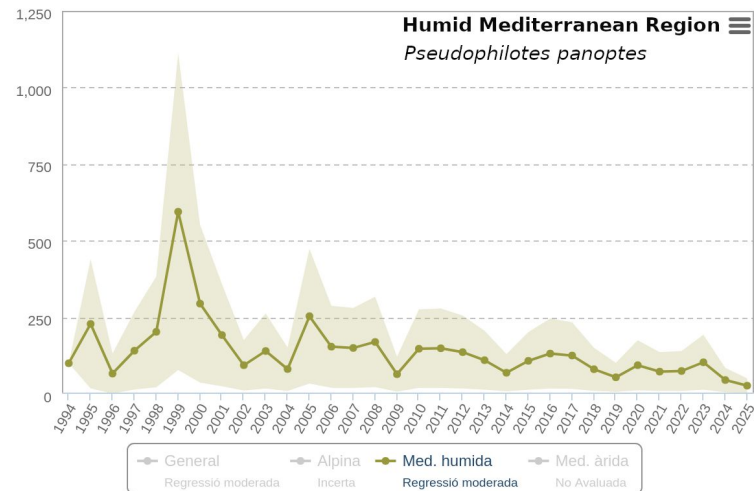
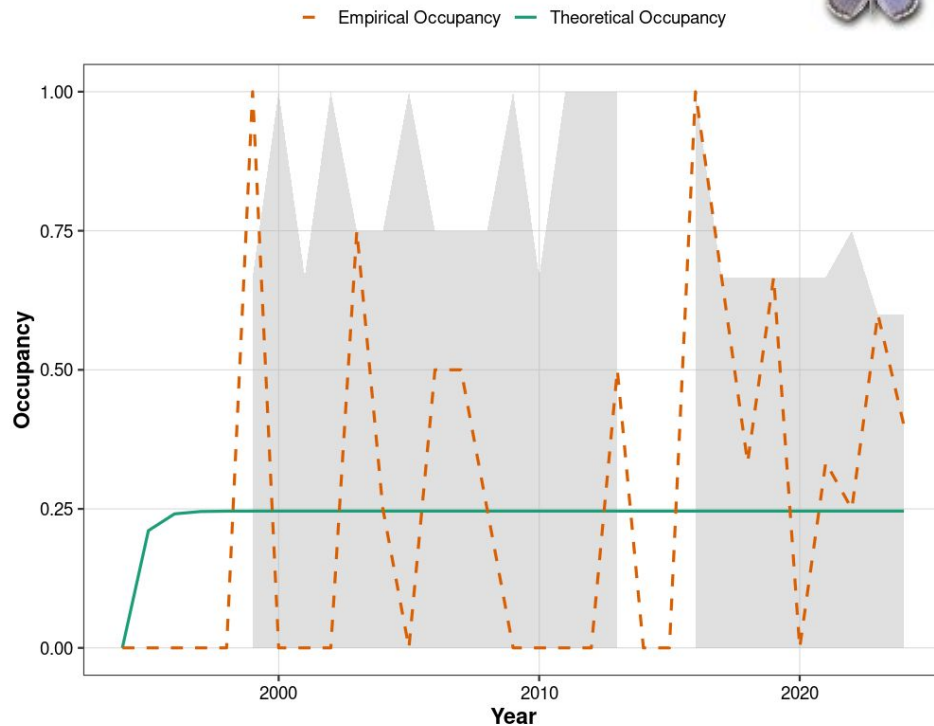


CBMS 2025



2. Temporal trends in occupancy

Pseudophilotes panoptes in arid mediterranean region



P-value: Significance test

- 1 Generate n replicates (data tables with the same structure as the data and the c and e of a real species).
- 2 For each replicate, estimate c and e and record the NLL
- 3 This creates a distribution of NLLs: $NLL_1, \dots, NLL_n$.
- 4 Use the observed NLL of the real data, $NLL^{(o)}$ to calculate the p – *valor*.
- 5 If p – *valor* > 0.95 , then reject the model.
- 6 Otherwise, admit that the structure of the true data cannot be distinguished from the model-generated synthetic data, and therefore that the model cannot be rejected.